

In [3]: `import pandas as pd`

```
df = pd.read_csv(r'C:\Users\Prashanth\Desktop\job\projects\EDA\customer_shopping_behavior.csv')
```

In [5]: `df.head()`

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Standard
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Neon
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Standard

In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
Customer ID          3900 non-null int64
Age                  3900 non-null int64
Gender               3900 non-null object
Item Purchased       3900 non-null object
Category             3900 non-null object
Purchase Amount (USD) 3900 non-null int64
Location             3900 non-null object
Size                 3900 non-null object
Color                3900 non-null object
Season               3900 non-null object
Review Rating        3863 non-null float64
Subscription Status  3900 non-null object
Shipping Type        3900 non-null object
Discount Applied     3900 non-null object
Promo Code Used      3900 non-null object
Previous Purchases   3900 non-null int64
Payment Method       3900 non-null object
Frequency of Purchases 3900 non-null object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
In [7]: # Summary statistics using .describe()
df.describe(include='all')
```

Out[7]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Su
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	
top	NaN	NaN	Male	Pants	Clothing	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	

```
In [8]: # Checking if missing data or null values are present in the dataset
```

```
df.isnull().sum()
```

```
Out[8]: Customer ID      0
Age      0
Gender    0
Item Purchased  0
Category  0
Purchase Amount (USD)  0
Location  0
Size      0
Color     0
Season    0
Review Rating  37
Subscription Status  0
Shipping Type  0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method  0
Frequency of Purchases  0
dtype: int64
```

```
In [9]: # Imputing missing values in Review Rating column with the median rating of the product category
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
In [10]: df.isnull().sum()
```

```
Out[10]: Customer ID      0
Age      0
Gender    0
Item Purchased  0
Category  0
Purchase Amount (USD)  0
Location  0
Size      0
Color     0
Season    0
Review Rating  0
Subscription Status  0
Shipping Type  0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method  0
Frequency of Purchases  0
dtype: int64
```

```
In [11]: # Renaming columns according to snake casing for better readability and documentation
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [12]: df.columns
```

```
Out[12]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
               'purchase_amount', 'location', 'size', 'color', 'season',
               'review_rating', 'subscription_status', 'shipping_type',
               'discount_applied', 'promo_code_used', 'previous_purchases',
               'payment_method', 'frequency_of_purchases'],
              dtype='object')
```

```
In [13]: # create a new column age_group
```

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
In [14]: df[['age', 'age_group']].head(10)
```

```
Out[14]:
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

```
In [15]: # create new column purchase_frequency_days
```

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
In [16]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

Out[16]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
In [17]: df[['discount_applied', 'promo_code_used']].head(10)
```

Out[17]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [18]: (df['discount_applied'] == df['promo_code_used']).all()
```

Out[18]: True

```
In [19]: # Dropping promo code used column
```

```
df = df.drop('promo_code_used', axis=1)
```

```
In [20]: df.columns
```

Out[20]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount', 'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 'shipping_type', 'discount_applied', 'previous_purchases', 'payment_method', 'frequency_of_purchases', 'age_group', 'purchase_frequency_days'], dtype='object')

```
In [21]: # connect with my sql
!pip install pymysql sqlalchemy
```

```
Collecting pymysql
  Downloading https://files.pythonhosted.org/packages/0c/94/e4181a1f6286f545507528c78016e00065ea913276888db2262507693ce5/PyMySQL-1.1.1-py3-none-any.whl (https://files.pythonhosted.org/packages/0c/94/e4181a1f6286f545507528c78016e00065ea913276888db2262507693ce5/PyMySQL-1.1.1-py3-none-any.whl) (44kB)
Requirement already satisfied: sqlalchemy in c:\programdata\anaconda3\lib\site-packages (1.3.9)
Installing collected packages: pymysql
Successfully installed pymysql-1.1.1
```

```
In [22]: from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "PK12345678"
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

# Write DataFrame to MySQL
table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

# Read back sample
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

Out[22]:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_ratin
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2

In []: